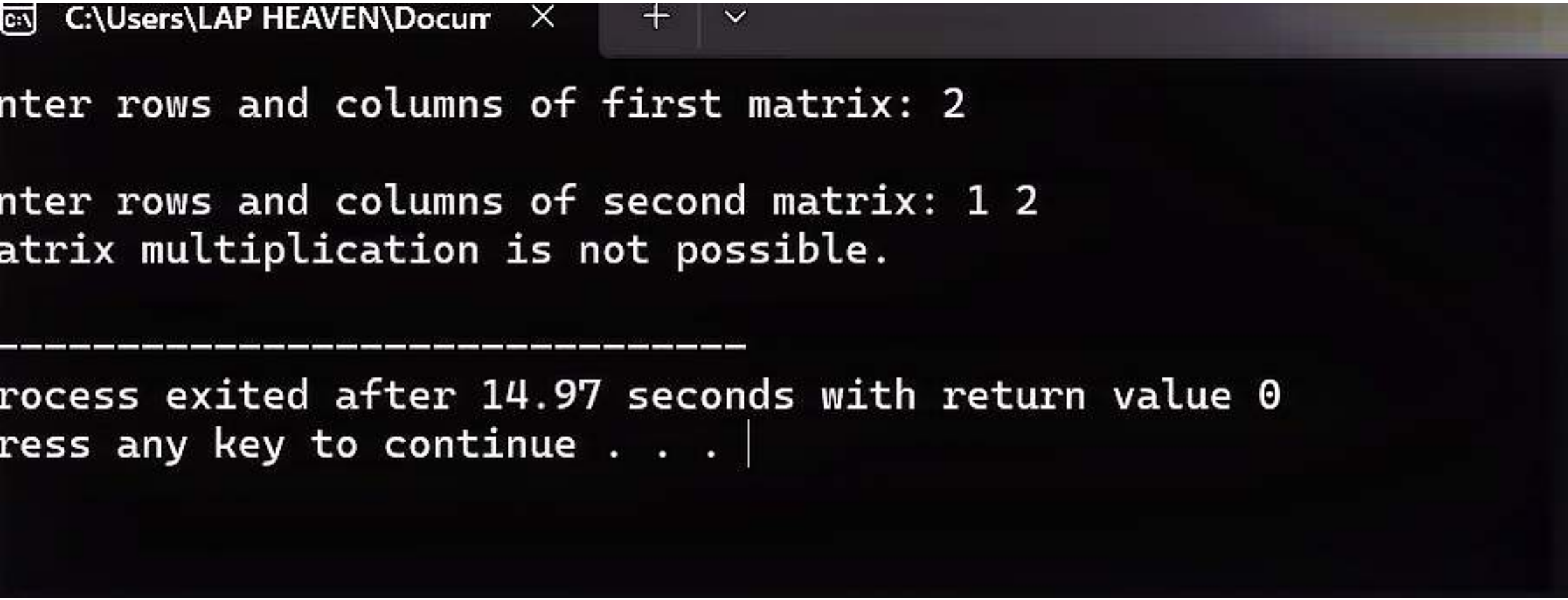

C:\Users\LAP HEAVEN\Docum
nter rows and columns of first matrix: 2
nter rows and columns of second matrix: 1 2
atrix multiplication is not possible.
rocess exited after 14.97 seconds with return value 0
ress any key to continue . . .

```c
#include <stdio.h>

int main()
{
    int a[3][3], b[3][3], result[3][3];
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d", &r2, &c2);

    if (r1 < 1 || r1 > 3 || c1 < 1 || c1 > 3 ||
        r2 < 1 || r2 > 3 || c2 < 1 || c2 > 3)
    {
        printf("Invalid matrix dimensions.\n");
        return 0;
    }

    if (c1 != r2)
    {
        printf("Matrix multiplication is not possible.\n");
        return 0;
    }

    printf("Enter first matrix:\n");

    for (i = 0; i < r1; i++)
```

```
    }
}

printf("\nFirst Matrix:\n");

for (i = 0; i < r1; i++)
{
    for (j = 0; j < c1; j++)
        printf("%d ", a[i][j]);
    printf("\n");
}

printf("\nSecond Matrix:\n");

for (i = 0; i < r2; i++)
{
    for (j = 0; j < c2; j++)
        printf("%d ", b[i][j]);
    printf("\n");
}

printf("\nResult of Multiplication:\n");

for (i = 0; i < r1; i++)
{
    for (j = 0; j < c2; j++)
        printf("%d ", result[i][j]);
    printf("\n");
}

return 0;
```